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Task: 8-puzzle problem

1. Problem Definition

The 8-puzzle is a state-space search problem in Artificial Intelligence (AI). It consists of a 3×3 board containing eight numbered tiles and one blank space. The objective is to transform the initial state into the given goal state by sliding a tile adjacent to the blank space into the blank position. The blank space can move up, down, left, or right, provided the move is within the board boundaries.

2. Initial State and Goal State

Initial State:

5 4 _
6 1 8
7 3 2

Goal State:

1 4 7
2 5 8
3 6 _

3. State-Space Representation

- Initial state: The given starting arrangement.
- Goal state: The required final arrangement.
- Operators: Move the blank space up, down, left, or right.
- Path cost: Each valid move has a cost of 1.

- Goal test: Check whether the current arrangement matches the goal state.

4. A* Search Algorithm

A* is an informed search algorithm that selects the most promising state using the evaluation function:

$$f(n) = g(n) + h(n)$$

Where:

- (g(n)): Actual cost from the initial state to the current state.
- (h(n)): Estimated cost from the current state to the goal state.
- (f(n)): Estimated total cost of the solution path through the current state.

The Manhattan distance heuristic is commonly used for the 8-puzzle. It calculates the sum of the horizontal and vertical distances of all numbered tiles from their goal positions, excluding the blank space.

$$h(n) = \sum_{i=1}^{8} (|x_i - x_i^*| + |y_i - y_i^*|)$$

Here, ((x_i,y_i)) is the current position of tile (i), and ((x_i^*,y_i^*)) is its goal position.

5. Manhattan Distance Calculation Table

The Manhattan distance for each tile in the initial state is calculated by comparing its current position with its goal position.

Tile	Initial position (row, column)	Goal position (row, column)	Manhattan distance
1	(2, 2)	(1, 1)	(|2-1|+|2-1|=2)
2	(3, 3)	(2, 1)	(|3-2|+|3-1|=3)
3	(3, 2)	(3, 1)	(|3-3|+|2-1|=1)

4	(1, 2)	(1, 2)	0
5	(1, 1)	(2, 2)	(|1-2|+|1-2|=2)
6	(2, 1)	(3, 2)	(|2-3|+|1-2|=2)
7	(3, 1)	(1, 3)	(|3-1|+|1-3|=4)
8	(2, 3)	(2, 3)	0
Total			14

Therefore, the initial heuristic value is:

$$h(n) = 2+ 3 + 1 + 0+ 2 + 2 + 4 + 0 = \boxed{14}$$

Since the initial state has not yet required any moves:

$$g(n) = 0$$

$$f(n) = g(n) + h(n) = 0 + 14 = \boxed{14}$$

6. Inversion Count and Solvability Calculation

An inversion occurs when a larger numbered tile appears before a smaller numbered tile in the row-wise sequence, ignoring the blank.
Initial state sequence: (5,4,6,1,8,7,3,2)

Tile	**Smaller tiles appearing after it**	**Number of inversions**
5	4, 1, 3, 2	4
4	1, 3, 2	3
6	1, 3, 2	3
1	None	0
8	7, 3, 2	3

7	3, 2	2
3	2	1
2	None	0
Total		16

Goal state sequence: (1,4,7,2,5,8,3,6)

Tile	Smaller tiles appearing after it	Number of inversions
1	None	0
4	2, 3	2
7	2, 5, 3, 6	4
2	None	0
5	3	1
8	3, 6	2
3	None	0
6	None	0
Total		9

For a 3×3 puzzle, the initial and goal configurations are mutually reachable only when their inversion counts have the same parity (both even or both odd).

- Initial inversion count = 16 (even).
- Goal inversion count = 9 (odd).

Therefore, the given initial state cannot reach the given goal state under the standard 8-puzzle rules. No sequence of legal moves can solve this exact configuration.

7. A* Search Algorithm Steps

1. Insert the initial state into the OPEN priority queue.
2. Calculate (g(n)), (h(n)), and (f(n)) for the initial state.

3. Select the state with the lowest (f(n)).
4. Check whether it matches the goal state.
5. If it does not, generate all valid successor states by moving the blank up, down, left, or right.
6. Calculate the evaluation function for each successor and update the OPEN list when a better path is found.
7. Maintain a CLOSED set to avoid repeatedly expanding previously explored states.
8. Continue until the goal is reached or OPEN becomes empty.

For the given configuration, A* search will not find a solution because the initial and goal states have different inversion-count parity.

8. Conclusion

The 8-puzzle demonstrates the application of state-space search and heuristic evaluation in Artificial Intelligence. The A* algorithm uses (f(n)=g(n)+h(n)), with Manhattan distance as a heuristic, to explore promising states efficiently. The initial state's Manhattan distance is 14. However, the inversion-count calculation shows that the initial state has 16 inversions while the goal state has 9. Since their parities differ, the puzzle as shown is unsolvable. A solvable version requires correcting either the initial or goal configuration so that both have the same inversion-count parity.